

Supplemental Material of “Spatio-Temporal Fusion and Adaptive Regularization Correlation Filter for Robust UAV Tracking”

Yang Zhang, Long Chen, *Senior Member, IEEE*, Zhengshan Wang, and Xuelin Xie

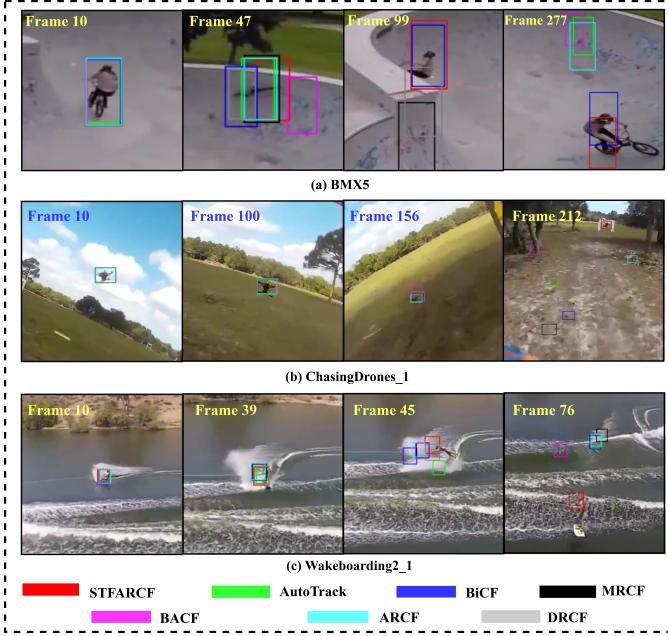


Fig. S1: Real-time tracking performance of STFARCF, AutoTrack, BiCF, MRCF, BACF, ARCF, and DRCF on the BMX5, ChasingDrones_1, and Wakeboarding2_1 video sequences.

To further evaluate the tracking performance of the proposed STFARCF tracker, we select six challenging UAV sequences—*BMX5*, *ChasingDrones_1*, *Wakeboarding2_1*, *person17_1*, *person21* and *uav3*—for real-time tracking analysis. STFARCF is compared against six representative trackers: AutoTrack, BiCF, MRCF, BACF, ARCF, and DRCF. As illustrated in Fig. S1(a), around frame 47, the target undergoes significant scale variation, resulting in drastic changes in appearance. This scenario demands strong adaptability in feature learning; several trackers begin to drift due to their inability to adapt to the evolving object representation.

This work has been supported by the Science and Technology Development Fund, Macao S.A.R. under Grant 0012/2025/RIA1 the University of Macau, the University of Macau Development Foundation under Grant MYRG-GRG2023-00106-FST-UMDF. (Corresponding Author: Long Chen).

Yang Zhang, Long Chen, Zhengshan Wang are with the Department of Computer and Information Science, Faculty of Science and Technology, University of Macau, Macau 999078, China (e-mail: yc47958@um.edu.mo; longchen@umac.mo; yc27436@umac.mo).

Xuelin Xie is with the School of Mathematics and Statistics, Wuhan University, Wuhan 430072, China, and also with the Faculty of Science and Technology, University of Macau, Macau 999078, China (e-mail: xl.xie@whu.edu.cn).

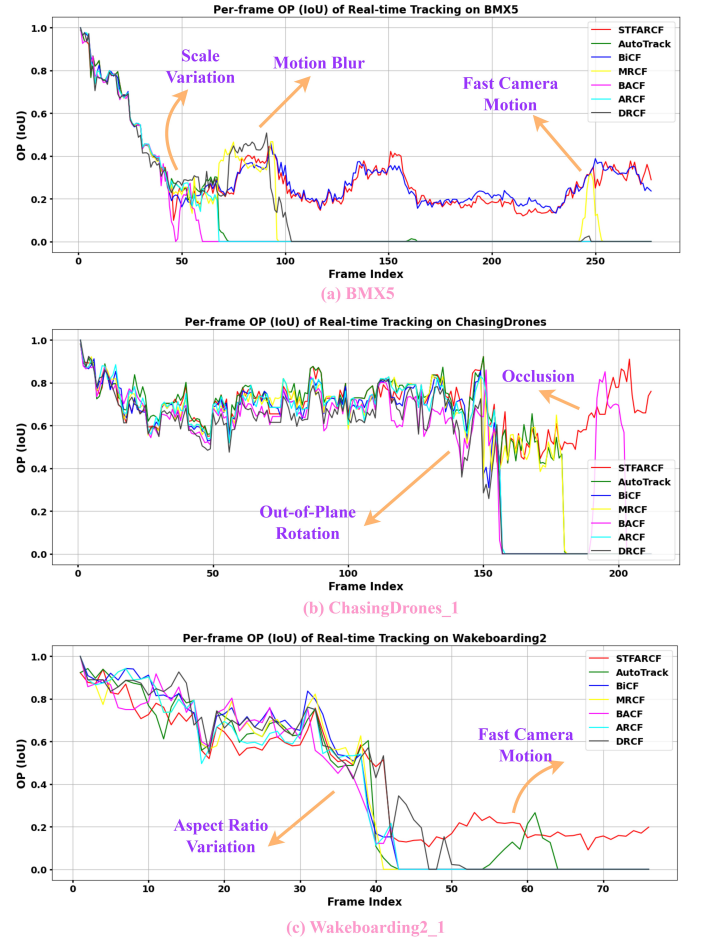


Fig. S2: Frame-wise real-time precision (Prec) performance of STFARCF, AutoTrack, BiCF, MRCF, BACF, ARCF, and DRCF on the BMX5, ChasingDrones_1, and Wakeboarding2_1 video sequences.

Subsequently, at approximately frame 99, the object experiences motion blur. In this case, only STFARCF and BiCF successfully maintain accurate tracking, while others fail to recover. Between frame 100 and frame 277, the sequence is characterized by fast camera motion. During this period, all compared trackers except STFARCF and BiCF fail to re-identify the object due to insufficient model update capabilities.

The frame-wise precision results are shown in Fig. S2(a). Around frame 47, all trackers exhibit a sharp drop in precision,

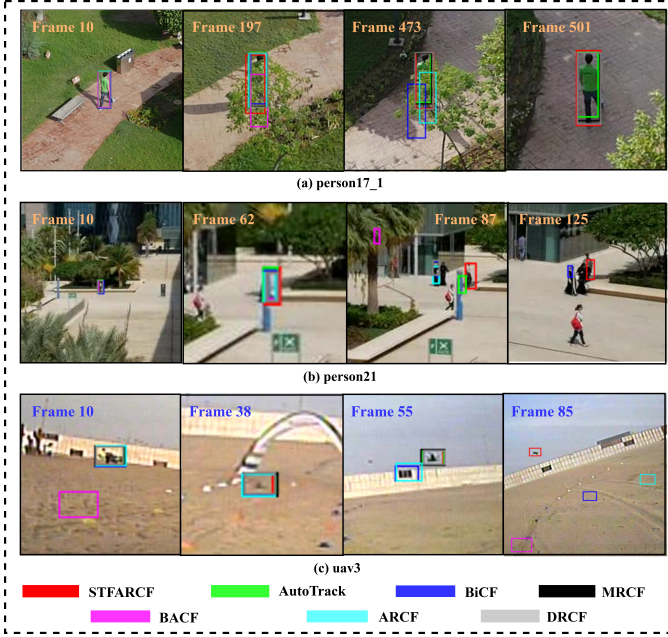


Fig. S3: Real-time tracking performance of STFARCF, AutoTrack, BiCF, MRCF, BACF, ARCF, and DRCF on the person17_1, person21, and uav3 video sequences.

with several degrading to zero. After recovering from this initial disturbance, STFARCF steadily improves its tracking accuracy. Although another temporary drop is observed around frame 99 due to motion blur, STFARCF quickly regains stability and maintains robust performance throughout the remainder of the sequence.

These results validate the effectiveness of the proposed ATRM and HDPP modules, which significantly enhance the tracker’s ability to perform rapid and reliable model updates under severe appearance changes caused by fast motion and frequent scale variations.

As shown in Fig. S1(b), around frame 150, the tracking scene undergoes an out-of-plane rotation during the drone chase sequence. Under such rotation-induced deformation, most trackers gradually lose the object due to their limited robustness to perspective changes. Furthermore, the small object size and occlusion introduce additional challenges. Between frames 150 and 200, only STFARCF and BACF manage to maintain tracking. However, due to BACF’s weak feature discriminability, it misidentifies a tree trunk—semantically similar to the object. Ultimately, only STFARCF succeeds in accurately tracking the object throughout the sequence.

The corresponding frame-wise precision is illustrated in Fig. S2(b). When out-of-plane rotation occurs, most trackers experience a sharp drop in precision, with many degrading to zero. In contrast, STFARCF maintains steadily increasing precision even during occlusion, demonstrating the effectiveness of the proposed STPSF module. By leveraging spatially aligned object fusion from multiple historical frames, STFARCF is able to preserve and reinforce the correct target representation. This enables reliable model updates despite partial feature loss caused by occlusion.

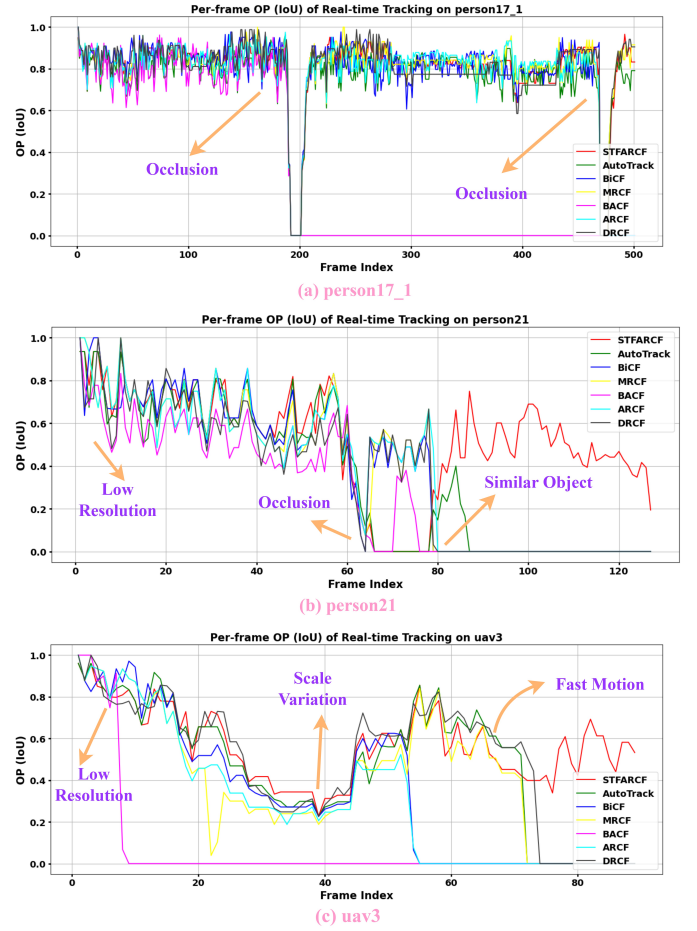


Fig. S4: Frame-wise real-time precision (Prec) performance of STFARCF, AutoTrack, BiCF, MRCF, BACF, ARCF, and DRCF on the person17_1, person21, and uav3 video sequences.

With the growing popularity of UAVs for capturing dynamic sports activities, Fig. S1(c) illustrates a real-time tracking scenario in which a drone autonomously follows a surfing athlete. During the surfing sequence, the athlete performs various actions such as jumps and flips, leading to frequent aspect ratio variations. Additionally, background clutter—such as splashing water—further complicates the scene. In this challenging setting, only the proposed STFARCF tracker is able to continuously and successfully follow the object, while all other trackers fail to maintain robust tracking.

Furthermore, when fast camera motion occurs later in the sequence, STFARCF continues to deliver stable and accurate tracking results. These observations demonstrate the superior robustness and tracking stability of STFARCF in highly dynamic and visually complex real-world environments, thereby validating its suitability for practical UAV-based applications.

Fig. S3(a) and Fig. S4(a) illustrate the real-time tracking performance of various trackers in a long-term sequence where the object undergoes two occlusion events. Benefiting from the proposed STPSF module, STFARCF effectively mitigates the negative impact of occlusion on tracking performance and successfully recovers the object after each occlusion.

Fig. S3(b) presents a scenario where the UAV tracks a small object from a long-distance aerial view, resulting in severe low-resolution conditions. Around frame 62, the object is fully occluded, and after reappearing, it is further challenged by the presence of a similar-looking distractor around frame 83. Although several trackers manage to re-lock onto the object after occlusion, they are misled by the similar distractor and fail to continue tracking the correct object. In contrast, STFARCF demonstrates superior feature learning and model update capability, effectively reducing the likelihood of filter degradation and distortion under such challenging visual conditions.

Finally, Fig. S3(c) highlights STFARCF's strong performance in tracking an extremely small and fast-moving object under low-resolution conditions. Even as the object size diminishes significantly, STFARCF continues to update the appearance model accurately, ensuring robust and precise tracking. These results validate the effectiveness of the proposed methods in handling occlusion, visual similarity, fast motion, and resolution degradation in UAV tracking scenarios.